Run report: a second invariant guess on the pair at $\ell = 8$

- **Study:** `research/dreg_invariant_pair_guess_20261029/`, pre-registered before any D3 run. No amendments.
- **Run:** 2026-10-29 22:07 UTC to 2026-10-30 01:29 UTC.
- **Class:** stage diagnostic. No `S`, no rho ratio, no speed claim.

What ran

- **Arm D3:** `x15 := x7 + c`, `x14 := x6 + f`, `x24 = g`, with 8 branches, on the 12 committed unsatisfiable (22, 8) draws.
  - It runs the same `dreg_linear_guess` build as 2851 and 2883, at `--to 4 --max-cols 140000 --check`.
  - Early stop at the first branch that saturates degree 4 without refuting.
- **Comparator S3:** the committed records of the two earlier studies.
- **Identity control N:** S3 on draw 5 branch 0, re-run.
- **Limits:** `prlimit --as=15e9`, `timeout 7200`, one process at a time, on the 4-core, 16 GB container, not isolated.

Completion

- 39 processes, all exited 0. None was killed or retried.
- Peak RSS was 7.0–8.5 GB.

Measured values

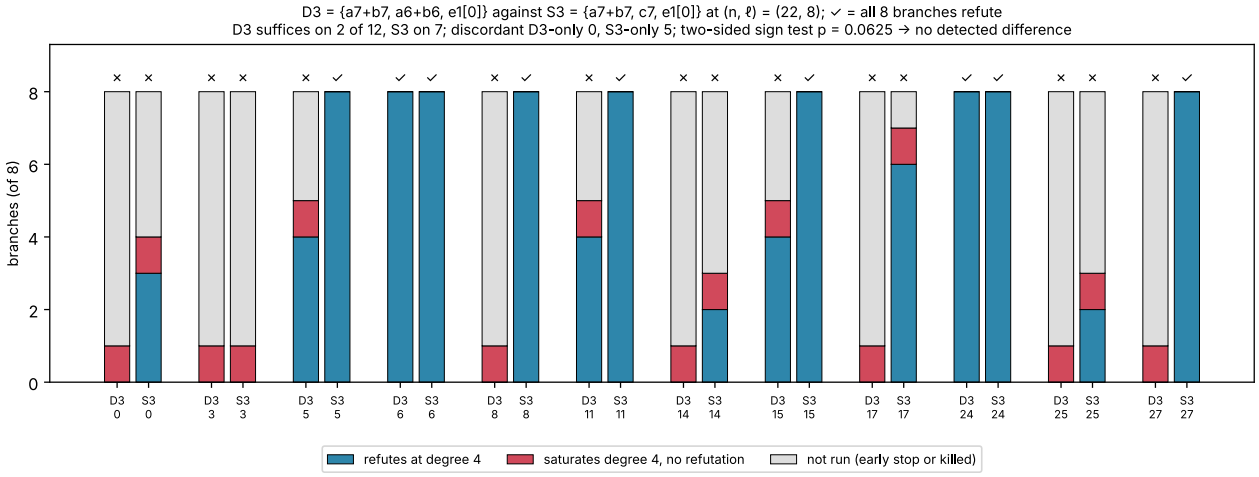


Figure 1: Branches per arm and draw: refuting at degree 4, saturating degree 4 without refutation, or not run. Source: `plot.py` (layout only) on `plot-data.json` from `dreg_linear_guess_pair_score --plot-data`.

- **Controls:** N matches the committed record, every record passes the identity and exactness checks, and every draw is decided. Valid.
- **D3** suffices on 2 of 12 draws, and **S3** on 7 of 12.
- **Discordant draws:** D3-only 0, S3-only 5.
- **Registered verdict:** no detected difference (two-sided exact sign test, p = 0.0625).
- D3 is never better on any draw.